



Advancing benthic diatom-based freshwater health assessment in China: from an index application to an integrated assessment framework

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Abstract Benthic diatoms are widely used as bioindicators in freshwater assessment, and diatom-based biomonitoring tools have been applied in China for several decades. However, existing knowledge in China remains fragmented across Chinese and English literature, geographical regions, ecosystem types, and methodological traditions. Here, we conducted a review of 62 studies that used benthic diatoms for freshwater bioassessment in China and summarized their geographical distribution, ecosystem coverage, publication trends, and bioassessment methods, and identified the future research needs in diatom monitoring. The review shows that selected studies are unevenly distributed across provinces and are dominated by riverine systems. Traditional, mainly European morphology-based indices remain central also in China, but localized and fine-tuned multimetric indices, molecular diatom indices, and environmental DNA-based approaches indicate a shift toward regional calibration and methodological integration while also highlighting the use of many

fit-for-purpose approaches (e.g., diatom teratological forms). We argue that the key challenge is no longer whether diatoms can indicate freshwater conditions in China, but how diatom-based assessment tools can be standardized, calibrated, and scalable across diverse freshwater ecosystems and biogeographical regions. Future progress requires development of standardized protocols and fit-for-purpose approaches, national taxonomic and molecular reference databases, long-term monitoring, and integration of emerging technologies.

Keywords Bioassessment · Biomonitoring · Ecological status · Diatom index · eDNA · River health

Introduction

Freshwater ecosystems in China are exposed to multiple and interacting pressures, including nutrient enrichment, chemical pollution, urbanization, habitat degradation, and climate-related disturbance (Dudgeon 2019; Chen et al., 2020; Xiong et al., 2026). Conventional physicochemical monitoring provides essential information on water quality, but it often reflects conditions at the time of sampling and may fail to capture cumulative ecological effects (Karr 1991; Vadas et al., 2022). Biological indicators integrate environmental conditions over biologically meaningful time scales, provide direct information on

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cumulative effects on multiple environmental pressures on biota, and therefore offer a more direct basis for assessing ecological conditions and ecosystem health (Birk et al., 2020).

Benthic diatoms are among the most widely used biological indicators in freshwater assessment (Pandey et al., 2017; Costa & Schneck 2022; Taurozzi et al., 2024). Their short generation time, broad distribution, wide autecological knowledge, high diversity, and sensitivity to nutrients, organic pollution, pH, conductivity, metals, and hydrological disturbance make them suitable for detecting changes in water quality and ecological condition (Soininen and Teittinen 2019; Smucker et al., 2024; Abubaker et al., 2026). In Europe, benthic diatoms are incorporated into biomonitoring under the Water Framework Directive, where the diatom composition occurring at study sites is one of the biological quality elements used for ecological status assessment (Masouras et al., 2021). In the United States, diatoms and periphyton have also been used in national and regional bioassessment programs to evaluate stream condition, nutrient enrichment, and anthropogenic disturbance (Potapova & Charles 2007). Existing reviews have summarized the ecological role of benthic diatoms, their responses to environmental gradients, and the development of diatom indices for water quality and river health assessment in Europe and the United States (Wu et al., 2017; Masouras et al., 2021).

In China, benthic diatom-based assessment has also developed over several decades. Earlier Chinese reviews introduced the indicator value of diatoms, summarized common diatom indices, and highlighted challenges such as taxonomic uncertainty, possible biogeographical factors affecting the inference (e.g., the low number of truly cosmopolitan diatoms), limited regional testing, and the need for a more complete biomonitoring system (Ran et al., 2015; Tan & Zhang 2018). More recent Chinese studies have applied established benthic diatom indices, developed locally calibrated diatom-based indices of biotic integrity (Liu et al., 2026), and explored molecular methods such as environmental DNA (eDNA) metabarcoding (Jiang et al., 2023). However, applying benthic diatom-based assessment across China remains challenging. Many widely used diatom indices were developed in Europe or other regions under specific taxonomic, ecological, biogeographical, and water-management contexts (Potapova & Charles 2007; Qu et al., 2014). These

indices provide useful starting points, but their species composition used when developing the index, ecological sensitivity values of the species, and class boundaries cannot be directly applied to Chinese freshwater ecosystems without validation and calibration (Qu et al., 2014). At the same time, China's freshwater ecosystems span broad climatic, geological, hydrological, and biogeographical gradients, which may influence diatom assemblages and index performance. Potential differences in species pools, natural water chemistry, pollution pressures, habitat conditions, and monitoring traditions may all influence diatom assemblages and index performance (Liu et al., 2022).

Despite the increasing application of benthic diatom-based assessment approaches in China, previous reviews on diatom-based biomonitoring have mainly provided narrative summaries of diatom ecology, index applications, and future directions (Tan & Zhang 2018; Xue et al., 2023; Liu et al., 2025). They have not quantitatively mapped how benthic diatom-based freshwater assessment has developed in China across Chinese and English literature, nor systematically compared its geographical distribution, ecosystem coverage, dominant metrics, and methodological changes through time. It remains unclear where Chinese diatom research has been concentrated, which freshwater ecosystems have received the most attention, which indices and metrics dominate current applications, and how the field has shifted from external index use toward regional calibration and methodological integration.

To address this gap, we conducted a systematic review of Chinese- and English-language studies on benthic diatom-based freshwater bioassessment in China. Specifically, we aim to (1) identify Chinese and English studies that used benthic diatoms for freshwater bioassessment in China; (2) quantify their geographical distribution, ecosystem coverage, and temporal patterns; (3) summarize the application of major benthic diatom-based indicators and assessment methods in China; and (4) discuss the challenges and future directions for benthic diatom-based freshwater bioassessment in China.

Materials and methods

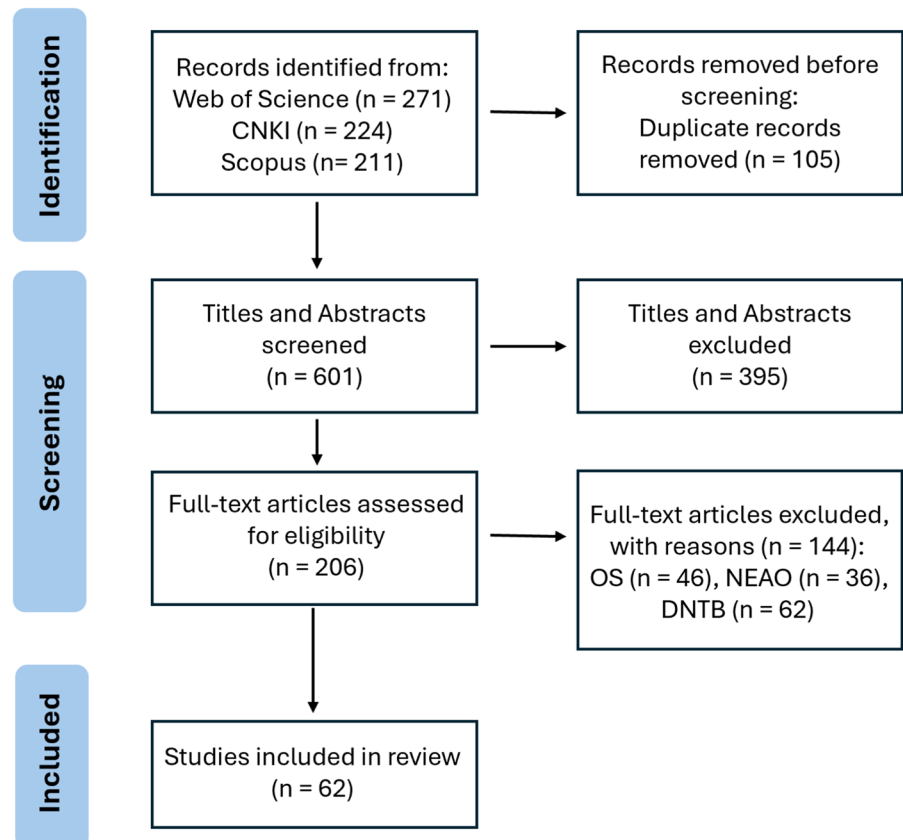
We conducted a literature search on the Web of Science (WOS), Scopus, and the China National

Knowledge Infrastructure database (CNKI) on August 31, 2026, with no restriction on publication year. The search strategy was designed to capture studies using benthic diatoms for freshwater bioassessment in China, including studies of periphytic, epilithic, epiphytic, and other substrate-associated diatom assemblages. Search terms combined benthic diatom terminology with freshwater ecosystem and assessment-related terms, including water quality assessment, ecological health assessment, trophic status evaluation, biological monitoring, and index-based bioassessment. Because search syntax and field codes differ among databases, database-specific search formats were used for WOS, Scopus, and CNKI. The detailed search strategy and full database-specific search strings are provided in Text S1.

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (O'Dea et al., 2021), which ensure a transparent and reproducible selection process (Fig. 1). After duplicate removal, titles and abstracts were screened to exclude records

clearly outside the scope of this review. Full texts of potentially relevant studies were then assessed using three inclusion criteria: (1) the study was conducted within China and the study ecosystem was a freshwater environment, including rivers, streams, lakes, reservoirs, wetlands, ponds, catchments, river–lake systems, or artificial waterways; (2) benthic diatoms were used as the target indicator group; and (3) the study used benthic diatom-based indicators, indices, metrics, or other diatom-derived tools to report explicit assessment outcomes, such as water quality grades, trophic status, ecological health levels, biological integrity scores, pollution levels, or other environmental assessment categories. Studies that only examined relationships between diatom assemblages or indices and environmental variables, stressor gradients, or habitat conditions, without reporting explicit assessment outcomes, were not considered eligible. Full-text articles were excluded for three main reasons: outside the scope (OS), diatoms not used as target bioindicators (DNTB), and no explicit assessment outcome (NEAO).

Fig. 1 Flow diagram showing the process of exclusion and inclusion of studies in the review based on the PRISMA protocol. Records were identified through searches in Web of Science, Scopus, and CNKI. The final dataset included studies that were conducted in China, focused on freshwater ecosystems, used benthic diatoms as target bioindicators, and reported an explicit assessment outcome. *OS* outside the scope, *NEAO* no explicit assessment outcome, *DNTB* diatoms not used as target bioindicators



For each included study, we manually checked the original article and extracted bibliographic and methodological information, including publication year, publication language, study province or region, freshwater ecosystem type, and benthic diatom-based assessment metrics. When a study involved multiple provinces, ecosystem types, or assessment metrics, all relevant categories were recorded. To facilitate comparison across studies, freshwater ecosystem types were grouped into two broad categories: river and stream systems, and lake and reservoir systems. Artificial waterways were grouped with riverine systems when the assessment context was primarily lotic or channel based. Assessment methods were extracted using the terminology reported in each paper. When different abbreviations or spelling variants referred to the same metric or index, they were standardized. The full names and abbreviations of all assessment metrics are provided in Table S1.

We summarized the selected studies in terms of their geographical distribution, ecosystem coverage, publication trends, and assessment metrics. Spatial distribution was mapped at the provincial level based on the study locations reported in each paper. Ecosystem coverage was summarized by counting the number of studies in each ecosystem category. Temporal patterns were analyzed using publication year and publication language. The frequency of assessment metrics was calculated after standardizing metric names and counting each metric once per study. To examine methodological development through time, we identified the first year in which each assessment metric or index appeared in the selected dataset. These years indicate the first appearance of a metric within the dataset used in this review and do not necessarily represent the first use of the metric in China.

Results and discussion

Spatial, ecosystem, and temporal distribution: uneven evidence

The literature screening resulted in a final dataset of 62 studies. The selected studies were unevenly distributed across China (Fig. 2A). Frequently represented provinces included Hubei ($n = 9$), Jiangxi ($n = 9$), Guangxi ($n = 8$), Heilongjiang ($n = 8$), Liaoning ($n = 8$), Zhejiang ($n = 7$), and

Guangdong ($n = 6$), while many provinces were represented by only one or two studies (Table S2). This spatial pattern suggests that current evidence is concentrated in several well-studied basins or regions, whereas large parts of China remain poorly represented in diatom-based freshwater assessment. Such unevenness matters because the transferability of indices and metrics cannot be evaluated reliably without broader geographic coverage.

Riverine systems dominated the selected literature (Fig. 2B). Of the 62 included studies, 53 investigated rivers and streams and 10 investigated lakes and reservoirs, with one study covering both ecosystem categories. Within the predefined scope of benthic diatom-based bioassessment, this pattern reflects the strong tradition of using benthic diatoms in lotic biomonitoring, where they are widely applied to assess nutrient enrichment, organic pollution, and other forms of ecological disturbance. In lentic systems such as lakes and reservoirs, bioassessment has often relied more strongly on phytoplankton-based approaches (Phillips et al., 2013), whereas benthic or periphytic diatom-based assessment frameworks remain comparatively less developed (but see e.g., Yang et al., 2020). This imbalance is important because lotic and lentic ecosystems differ in hydrology, residence time, substrate availability, nutrient dynamics, and dominant pressure gradients (Lu et al., 2022; Yan et al., 2024; Zuo et al., 2024). A useful example is provided by Yu et al., (2022) who developed the Comprehensive Diatom Index (CDI) using periphytic diatom data from both lower Yangtze lakes and the Qingyi River system. The CDI responded to water quality gradients in both ecosystem types, but different assessment criteria were established for rivers and lakes. This suggests that some benthic diatom indices may be applicable across both lotic and lentic systems, but their transferability still needs to be tested under different ecosystem conditions.

The temporal distribution of the selected studies showed that publications were sporadic before 2010 but became more frequent from the early 2010s onward (Fig. 2C). Annual publication numbers subsequently fluctuated, with studies appearing in both Chinese- and English-language literature. Overall, 34 of the 62 included studies were published in Chinese and 28 in English.

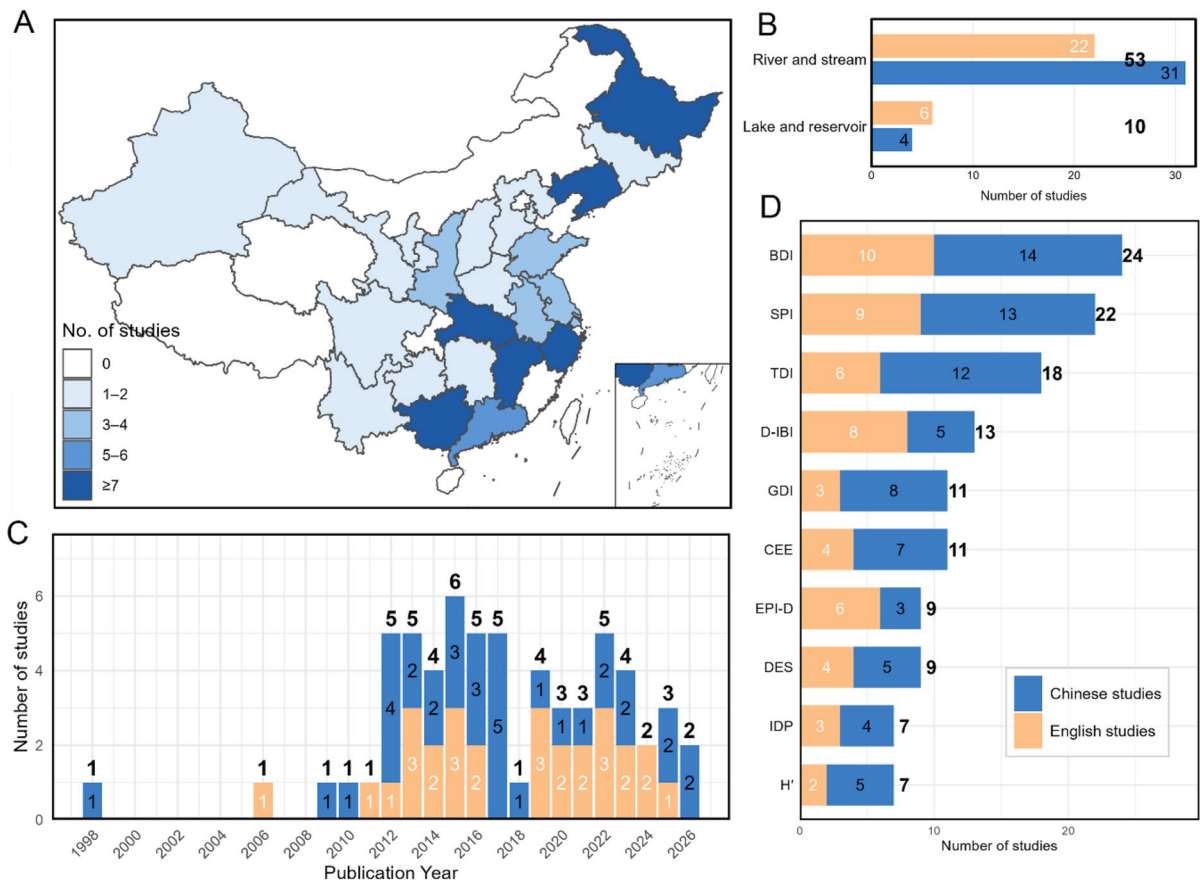


Fig. 2 Evidence mapping of selected Chinese and English studies on benthic diatom-based freshwater bioassessment in China. Panels summarize the **A** spatial distribution of studies,

B ecosystem types, **C** time trend, and **D** frequency of top 10 assessment metrics. Abbreviations of all assessment metrics are provided in Table S1

Dominant metrics: from non-local indices to localized multimetric assessment

After standardizing metric names, Biological Diatom Index (BDI), Specific Pollution Sensitivity Index (SPI), Trophic Diatom Index (TDI), Diatom-based Index of Biotic Integrity (D-IBI), European Economic Community Index (CEE), and Generic Diatom Index (GDI) were among the most common diatom indices (Fig. 2D), many of which originated from European assessment systems. Among these, BDI, SPI, TDI, CEE, and GDI are examples of the traditional single-index-based approach summarized in Table 1, whereas D-IBI represents a multimetric approach. Diversity and community metrics such as Shannon, richness, Margalef, Simpson, and Pielou evenness were also used (Table S1).

The frequent use of indices developed outside China indicates that benthic diatom-based assessment in China has been strongly influenced by established European and other regional biomonitoring systems. These indices are well established and widely used, and provide practical links between diatom assemblages and environmental gradients. However, their dominance also highlights the central issue of transferability. Indices developed under particular taxonomic, ecological, and regulatory contexts may not perform consistently under Chinese conditions. Their species lists, ecological sensitivity values, and class boundaries require local validation before they can be used for management decisions (Xue et al., 2023). For example, Soininen et al., (2019) & Pajunen et al., (2020) showed that diatoms do have different ecological preferences in different biogeographical

Table 1 Major methodological approaches for benthic diatom-based freshwater bioassessment in China: strengths, limitations, suitability, and future needs.

Method	Main advantages	Main limitations	Suitability for Chinese freshwater assessment	Future improvement needs
Traditional single-index-based approach (e.g., BDI, SPI, TDI, CEE, GDI)	Well established, widely used, and comparable with international studies; effective for detecting nutrient enrichment and organic pollution	Time-consuming; requires taxonomic expertise; affected by observer bias and incomplete species coverage in foreign index databases. Transferability may be questionable	Useful as a baseline method, especially when validated against local environmental and biogeographical gradients	Standardized microscopy protocols, quality control, and specific taxonomic databases for China. Ecological class boundaries for index interpretation should be regionally calibrated and validated
Multimetric approach (e.g., D-IBI, D-MMI)	Integrates multiple ecological attributes and can be calibrated to local freshwater conditions	Requires reliable reference sites, sufficient sample sizes, and local threshold setting; transferability may be limited	Strong potential and scalability for basin-scale health assessment in China	Harmonized metric selection, reference-condition criteria, and validation across basins
Trait-based approach	Provides functional interpretation and may reduce dependence on exact species identity	Trait databases for Chinese diatoms remain incomplete; trait responses may vary by region	Promising for improving ecological interpretation and cross-basin comparison	Develop specific trait databases for China and validate trait responses to key stressors
eDNA metabarcoding	Efficient, high-throughput, and can reduce dependence on expert microscopy	Species-level identification and abundance estimation remain uncertain; reference libraries may be incomplete or taxonomically inconsistent	Promising as a complementary tool for routine monitoring and taxonomic consistency	Build curated regional molecular reference libraries linked to morphologically validated reference material and harmonized taxonomy, and standardize markers and bioinformatic workflows
Automated image analysis/AI-assisted identification	May improve identification rate, consistency, and large-scale monitoring capacity	Requires large annotated image datasets and expert validation; rare taxa remain challenging	Emerging but potentially useful for future national monitoring programs	Build open image libraries and train models using Chinese diatom taxa
Integrated assessment framework	Combines morphology, indices, traits, eDNA, AI, and environmental data for stronger interpretation	More data-demanding and requires coordination across methods and agencies	Best long-term direction for a scalable national freshwater assessment framework	National standardization, inter-basin calibration, long-term monitoring, and shared databases

regions and diatom indices may not be transferrable even to closely located countries.

Representative Chinese studies support this interpretation. Studies in the upper Han River showed that some indices were related to trophic status and ionic conditions, but their performance differed between dry and wet seasons (Tan et al., 2013). In the Taizi River, European diatom indices responded significantly to nutrient and organic pollution gradients, but their classifications did not correspond directly to Chinese surface water quality classes (Qu et al., 2014). This mismatch likely reflects differences between the two assessment systems, as the Chinese standard is based mainly on designated water uses and physicochemical thresholds, whereas diatom indices are based on biological responses to environmental conditions (Fan et al., 2015). Studies in the Tangwang River and Longjiang River further demonstrated that different indices may perform differently depending on local environmental gradients (Tang et al., 2014; Xue et al., 2023), selected index systems, and analytical approaches. A more explicit example of regional calibration is the development of the Yuqiao Reservoir Watershed Diatom Assemblage Index (YRDAI), which localized the Japanese Diatom Assemblage Index to organic water pollution (DAI_{po}) by reclassifying diatom taxa according to their relationships with regional pollution gradients (Liu et al., 2022). In that study, the correlation-based YRDAI better reflected chemical oxygen demand and total phosphorus pollution than the original DAI_{po}, showing that borrowed indices can be improved when local species–environment relationships are incorporated. These cases show that external indices can be useful, but they should be treated as transferable hypotheses rather than universally applicable tools.

Multimetric and localized approaches have become increasingly important, with D-IBI representing a key step from single-index application toward locally calibrated ecological health assessment in China (Tan et al., 2015; Liu et al., 2026). Compared with single-purpose diatom indices, D-IBI frameworks integrate multiple dimensions of community response, including taxonomic diversity, species composition, pollution tolerance, habitat-related attributes, and, increasingly, functional metrics. For example, a D-IBI developed in the Thousand Islands Lake catchment incorporated functional metrics into the candidate metric pool and finally selected four metrics,

including species richness, Shannon diversity, functional divergence, and functional redundancy (Wang et al., 2023). This shows that D-IBI is well beyond taxonomic composition and better reflects how external disturbance affects both community structure and ecosystem function. However, localized D-IBI frameworks also create new challenges. Reference-site selection varies among regions and often depends on local environmental conditions, available monitoring data, and expert judgement. At present, there is no unified national standard for defining reference sites, selecting candidate metrics, or setting scoring thresholds. Therefore, a D-IBI developed for one basin may improve local diagnostic accuracy, but it does not automatically solve the problem of cross-basin or national comparability.

Method development in time: from European indices to methodological integration

The temporal pattern of the selected studies suggests a gradual methodological transition (Fig. 3). The earliest records in the selected dataset used established diatom indices such as DAI_{po} and River Pollution Index (RPID) in 1998, followed by River Diatom Index (RDI) in 2006 and Biological Diatom Index (BDI) and SPI in 2009. From the early 2010s onwards, studies increasingly applied a broader set of conventional diatom indices, particularly trophic, saprobic, and pollution-sensitivity indices such as TDI, CEE, and GDI, together with community metrics such as Shannon diversity, Simpson diversity, and Margalef richness. This indicates that Chinese diatom-based assessment was initially influenced by imported or established index systems, while also incorporating general community-based metrics. More localized and multimetric approaches subsequently became more visible in the dataset. D-IBI appeared in 2011, followed by Diatom Multimetric Index (D-MMI) in 2019, and YRDAI in 2022, suggesting a shift from applying single indices toward developing regionally calibrated or integrated assessment frameworks. Molecular approaches have also begun to appear in Chinese diatom research. For example, Jiang et al., (2023) used eDNA metabarcoding in the Shaying River basin to compare operational taxonomic unit (OTU)- and amplicon sequence variant (ASV)-based, taxonomy-assigned and taxonomy-free molecular data strategies, and

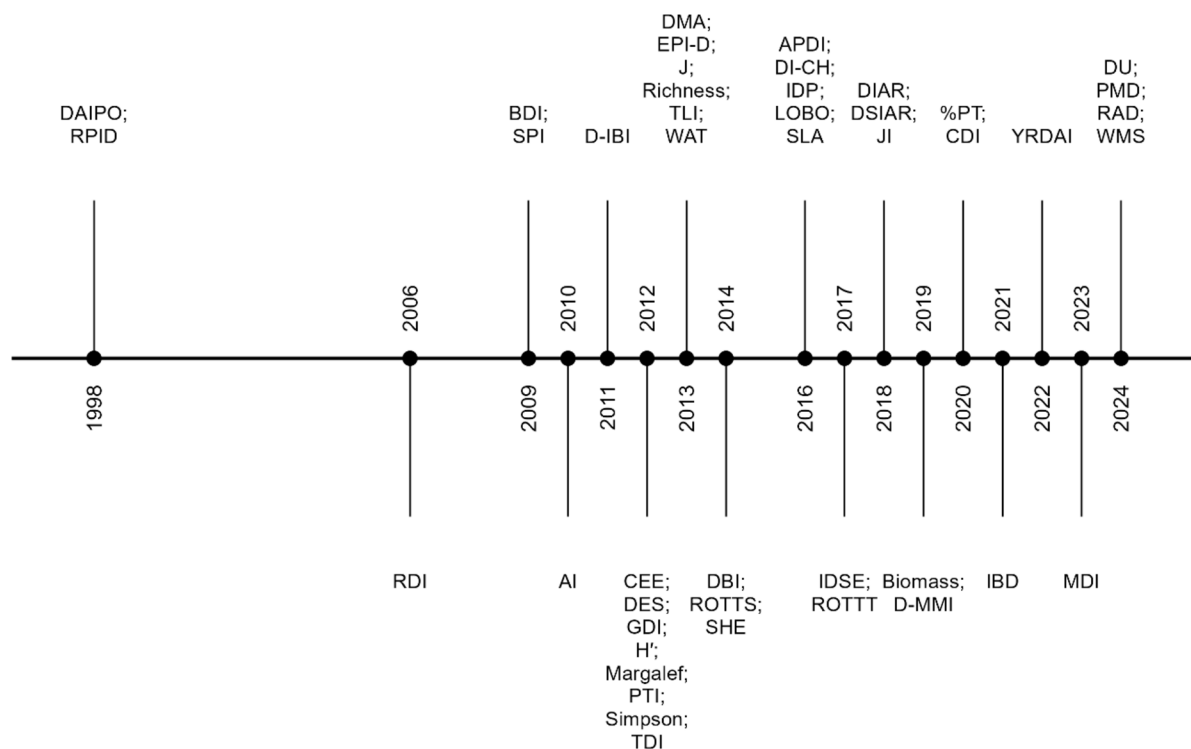


Fig. 3 Timeline of the first appearance of benthic diatom-based assessment metrics in the selected dataset. The timeline shows the first year in which each benthic diatom-based assessment metric or index appeared among the selected Chinese- and English-language studies on freshwater

bioassessment in China. Years indicate the first appearance of each metric within the selected dataset, and do not necessarily represent the first use of the metric in China. Abbreviations of all assessment metrics are provided in Table S1

developed a molecular diatom index (MDI). This illustrates the potential of eDNA-based approaches for future river health assessment and highlights the need to standardize bioinformatic workflows and molecular index calculation.

This temporal pattern shows that diatom-based assessment in China has not followed a single methodological pathway. Instead, the field has expanded from the application of established diatom indices to a broader set of localized, multimetric, molecular, and automated approaches. To clarify the role of these approaches in future assessment frameworks, we summarize the main advantages, limitations, suitability, and future needs of major method types in Table 1. Overall, no single method can address all assessment needs. Imported diatom indices remain useful for rapid screening and international comparison, but they require regional validation before being used for management decisions (Qu et al., 2014). Localized multimetric indices improve regional relevance and

ecological interpretation, but depend on reference-site selection, sufficient sampling effort, and harmonized metric scoring. Trait-based approaches can provide functional interpretation (Riato et al., 2022), whereas eDNA and artificial intelligence (AI)-assisted tools may improve monitoring efficiency and taxonomic consistency (Wang et al., 2024), but they depend on well-developed taxonomic, molecular, and image reference databases.

Challenges and future directions

Regional differences and methodological transition

The evidence reviewed above suggests that the application of benthic diatom-based assessment in China is shaped by two linked challenges. The first is external index transferability. Many commonly used indices were developed outside China,

especially in Europe, where monitoring systems are supported by long-term datasets, reference-condition approaches, and intercalibrated ecological status classes (Masouras et al., 2021). In China, these indices can detect broad pollution or trophic gradients, but their sensitivity values and class boundaries may not match local water chemistry, species pools, or management standards (Jin et al., 2025).

The second challenge is high level of environmental variation within China. Chinese freshwater ecosystems span large climatic, geological, hydrological, and biogeographic gradients. Human pressures also vary among regions, including agricultural nutrient enrichment, urban runoff, industrial pollution, hydrological alteration, and habitat degradation. These differences shape diatom communities and determine which indicators respond most strongly to environmental stress. In addition, methodological inconsistency among studies, such

as differences in sampling season, substrate type, taxonomic resolution, reference-site selection, and index calculation, can further reduce comparability (Liu et al., 2024).

These challenges mean that China does not need to choose simply between imported and local indices. Instead, diatom-based assessment should move through a staged methodological transition: from the application of imported indices, to regional validation, to localized multimetric assessment, and finally to the integration of emerging tools such as eDNA, trait-based metrics, automated image analysis, and AI-assisted identification (Fig. 4). This transition requires both regional calibration and national coordination.

Future opportunities and target framework

Based on the evidence mapping, we propose that diatom-based freshwater assessment in China should

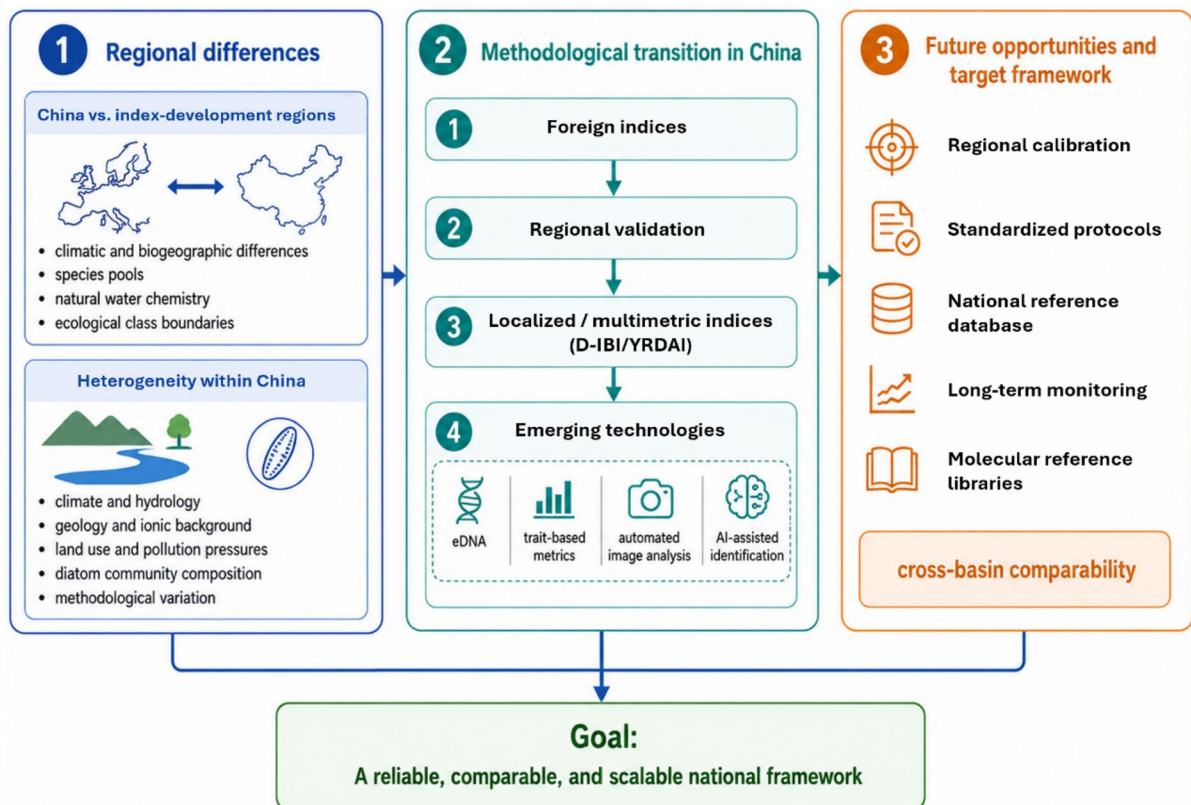


Fig. 4 Conceptual framework for advancing benthic diatom-based freshwater bioassessment in China. The framework illustrates a transition from external index application toward an integrated assessment framework

move from case-specific index application toward an integrated and operational assessment framework (Fig. 4). This framework should include five connected components.

First, benthic diatom indices used in China should be regionally localized and calibrated. Existing indices can remain useful tools, but their performance should be tested against local environmental gradients, species pools, ecosystem types, and management categories. The YRDAI case illustrates one practical pathway: an existing index can be localized by reclassifying taxa according to regional pollutant gradients and then comparing the localized index with the borrowed version (Liu et al., 2022). For lentic ecosystems, especially lakes and reservoirs, benthic diatom-based assessment remains relatively underdeveloped, while ecological assessment has more commonly relied on phytoplankton-based approaches. If benthic diatoms are to be used more widely in these systems, indices need to be developed or calibrated using data from lakes and reservoirs and validated against environmental gradients and habitat conditions specific to standing waters. For example, Yu et al., (2022) developed the Comprehensive Diatom Index (CDI) using periphytic diatom data from both lower Yangtze lakes and the Qingyi River system. The CDI responded to water quality gradients in both ecosystem types, although different assessment criteria were used for rivers and lakes. Such studies can help determine which components of existing indices are transferable across ecosystem types and which require recalibration for lentic environments.

Second, standardized protocols are needed for sampling, slide preparation, taxonomic identification, metric calculation, scoring, and ecological status classification. The national “Technical guidelines for water ecological monitoring—aquatic organism monitoring and evaluation of rivers” in China (Ministry of Ecology and Environment of the People’s Republic of China 2023) provides a standardized framework for river aquatic organism monitoring and assessment, including site selection, monitoring frequency, sampling methods, quality assurance and quality control, and assessment methods. It also includes periphytic algae and provides procedures for the comprehensive diatom index (CDI), together with reference values for ecological classification. This represents an important step toward standardized river biomonitoring in China. Under the European Water

Framework Directive, observed biotic communities are compared with type-specific reference conditions representing minimal human disturbance. This requires defining reference expectations for different river and lake types, which can be difficult where suitable reference sites are scarce (Birk et al., 2012). Nonetheless, such an approach could also be considered more broadly in China (Poikane et al., 2014).

Third, China needs national taxonomic, ecological, trait, image, and molecular reference databases for diatoms. In particular, molecular reference libraries should be developed in close connection with traditional morphotaxonomy. Comparisons between morphology- and metabarcoding-based diatom assessment have shown that incomplete reference libraries and taxonomic inconsistencies can lead to substantial differences in taxonomic inventories and may affect downstream index calculation (Vasselon et al., 2017; Pawlowski et al., 2018). Reliable molecular reference libraries should therefore link commonly used barcode markers, such as *rbcL* and *18S*, to morphologically validated reference material, including cultured strains and, where possible, archived voucher material, while maintaining harmonized nomenclature and traceable taxonomic information as classifications are revised (Leese et al., 2018). Such taxonomic–molecular coupling would improve the reliability of sequence assignment and help connect molecular indices with the ecological knowledge accumulated through morphology-based biomonitoring. Trait-based and AI-assisted approaches would similarly benefit from linking trait and image records to the same taxonomic backbone. Without these interconnected reference infrastructures, emerging tools may improve efficiency but still produce uncertain or poorly comparable assessment outcomes (Apothéoz-Perret-Gentil et al., 2021).

Fourth, long-term and multi-season monitoring should be strengthened. Many diatom studies in China remain spatially or temporally limited, making it difficult to distinguish pollution gradients from seasonal, hydrological, or interannual variation. Long-term monitoring would also support the development of reference expectations and improve the detection of ecological change, degradation, and recovery (Revenge et al., 2005).

Finally, different methodological layers should be combined rather than treated as competing

alternatives. Morphology-based indices provide continuity with existing monitoring. Localized multimetric indices improve regional ecological interpretation. Trait-based and multivariate approaches can help reveal mechanisms of community response. eDNA and AI-assisted tools may improve efficiency, reproducibility, and taxonomic consistency, but they require validation and standardization before routine use (Venkataramanan et al., 2024; Wang et al., 2024). This aligns with recent freshwater biomonitoring perspectives, which emphasize that emerging technologies should be integrated carefully with existing approaches rather than replacing them without calibration.

The goal is therefore not to develop a single diatom index to be applied uniformly across all Chinese freshwater ecosystems. Rather, the goal should be a reliable, comparable, and scalable national framework that allows regional and ecosystem-specific calibration, while maintaining standardized protocols, shared databases, quality control, and cross-basin comparability. Such a framework would also allow diatom-based assessment to be integrated with broader aquatic bioindicator systems, including macroinvertebrates, fish, macrophytes, phytoplankton, and microbial indicators.

Conclusions

This review used systematic literature screening and evidence mapping to synthesize the development of benthic diatom-based freshwater bioassessment in China. The selected studies show a clear methodological broadening, from traditional diversity metrics and imported diatom indices toward regional validation, localized multimetric indices, and emerging molecular, trait-based, and automated approaches. However, morphology-based indices remain dominant, and research coverage is still uneven across provinces and ecosystem types.

The central challenge is not whether diatoms can indicate freshwater condition, but how their use can be standardized, calibrated, and integrated across diverse Chinese freshwater ecosystems. Foreign, mainly European indices, provide valuable starting points, but require validation under Chinese conditions. Localized multimetric frameworks improve regional relevance, but need harmonization and

cross-system testing. Emerging tools such as eDNA metabarcoding, trait-based metrics, automated image analysis, and AI-assisted identification offer new opportunities, but depend on high-quality reference databases and standardized workflows.

Future diatom-based freshwater assessment in China should therefore move toward an integrated framework that combines regional calibration, standardized protocols, national taxonomic and molecular reference databases, ecological class boundaries for index interpretation, long-term monitoring, and complementary morphological, molecular, functional, and automated approaches. Such a framework would improve the reliability, comparability, and management relevance of diatom-based bioassessment, supporting freshwater ecosystem protection, restoration prioritization, and water ecological security in China.

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Author contributions Guohao Liu: Conceptualization, Data curation, Formal analysis, Writing—original draft, and Writing—review and editing. Janne Soininen: Supervision, and Writing—review and editing (lead).

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Data availability Data will be made available on reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

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